

CS1010 Discrete Mathematics for Computer Science

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Introduction to Cryptography

What is Cryptology?

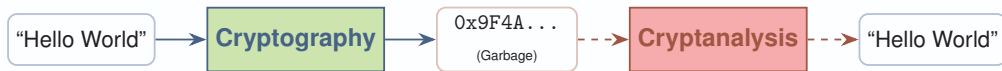
- ▶ **Cryptography:** The art of writing in **secret characters**.
 - ▷ Greek word “krptos” meaning hidden, and “graphia”, meaning writing.
- ▶ **Cryptanalysis:** The analysis and deciphering of **secret writings**.
- ▶ **Cryptology:** The scientific study of both of the above.



In our context, **cryptology** is the scientific study of the protection of information.

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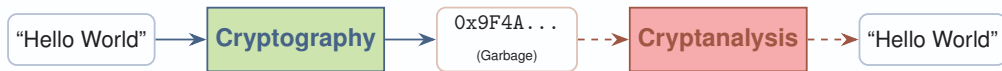
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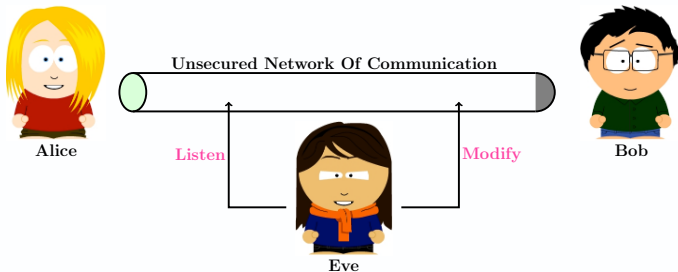
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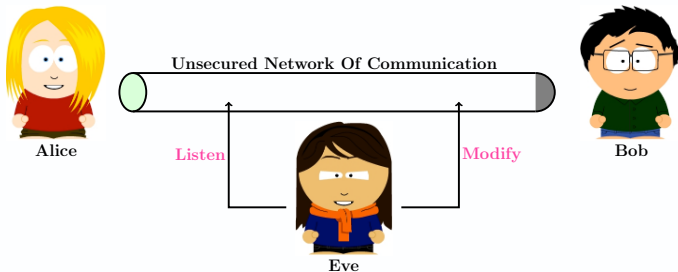
Why Cryptography?



The Goal is to develop **Efficient** algorithms with the following functionalities:

- ▶ **Confidentiality:** Prevents unauthorized disclosure of information.
- ▶ **Integrity:** Ensuring data has not been modified without detection.
- ▶ **Authenticity:** Ensures that the recipient can verify the identity of the sender.

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Applications of Cryptography

1. **Secure communication**
(encryption, authentication, messaging, email)
2. **Data security**
(files, databases, cloud storage)
3. **Authentication and access control**
(passwords, PINs, MFA)
4. **Digital identity**
(secure identification, certificates)
5. **Electronic commerce**
(online shopping, secure payments)
6. **Banking and digital payments**
(electronic funds transfer, digital cash)
7. **Blockchain and cryptocurrencies**
8. **Electronic voting**
9. **Digital signatures and contract signing**
10. **Copyright and digital rights management**
11. **Secret sharing**
12. **Zero-knowledge proofs**
13. **Secure multi-party protocols**
14. **Many more emerging applications**

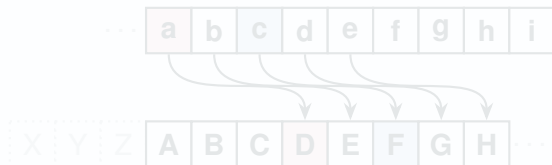


Classical Ciphers

Caesar's Cipher From History

- ▶ The first known **substitution cipher**, designed by Julius Caesar (58-44 BCE)
- ▶ Each character of a message is replaced by a character **three** position down
- ▶ $C = E(P) = (P + 3) \pmod{26}$ and $P = D(C) = (C - 3) \pmod{26}$

Plaintext: a b c d e f g h i j k l m n o p q r s t u v w x y z
Ciphertext: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C



Example:

hello → **KHOOR**

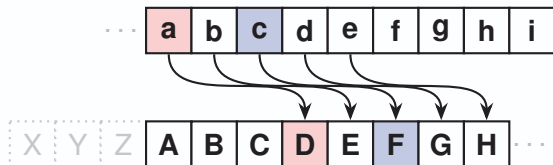
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Everyone who knows the encryption scheme can decrypt any message.

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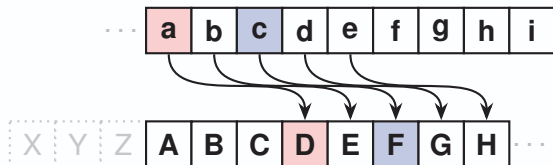
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Shift Ciphers

When Augustus came to power, the cipher was changed to a shift of one letters.

The Cryptographic Key

- ▶ K : The secret key known **only** to the sender and the receiver
- ▶ $C = E(K, P) = (P + K) \bmod 26$ and $P = D(K, C) = (C - K) \bmod 26$
- ▶ Generalized Caesar cipher, $0 \leq K \leq 25$

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Permutation Ciphers

The letters are permuted by some secret rule.

- ▶ Caesar's cipher is a permutation cipher, with 26 possible permutations
- ▶ Replace the simple cyclic permutation with a **any random** permutation
- ▶ The total number of such random permutations is enormous: $26! \approx 4 \cdot 10^{26}$
- ▶ These are called **Monoalphabetic Substitution Ciphers**.
- ▶ Such ciphers were used extensively during the end of 19th century

Monoalphabetic Substitution Example

Example of Secret Key

Plaintext: a b c d e f g h i j k l m n o p q r s t u v w x y z
Ciphertext: Q A Z X S W E D C V F R T G B N H Y U J M K I L O P

Encryption Process:

Plaintext:

cryptographicprotocol

Ciphertext:

ZYONJB EYQNDCZNYBJBZBR

Decryption Process:

Ciphertext:

ZYONJB EYQNDCZNYBJBZBR

Plaintext:

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Security Evaluation

The size of the key space is: $26! \approx 4 \cdot 10^{26} \approx 1.3 \cdot 2^{88} \approx 89$ bits

- ▶ Impractical to search this entire key space exhaustively via brute-force
- ▶ The probability of guessing the key is extremely small.0

Are there algorithmic shortcuts that can help an attacker break it?

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A Simple Ciphertext-Only Attack

- ▶ Monoalphabetic substitution ciphers are vulnerable if the attacker knows the structural language distribution of the underlying plaintext (e.g., standard English text)
- ▶ The statistical distribution of letters is **invariant to permutation**
- ▶ Each letter is substituted to a new character, but that new character inherits the exact frequency weight of the original letter
- ▶ For example, the most frequent letter in standard English text is **e** ($\approx 12.31\%$).

Letter	Freq
e	12.31%
t	9.59%
a	8.05%
o	7.94%
n	7.19%

Letter	Freq
i	7.18%
s	6.59%
r	6.03%
h	5.14%
l	4.03%

Letter	Freq
d	3.65%
c	3.20%
u	3.10%
m	2.25%
f	2.28%



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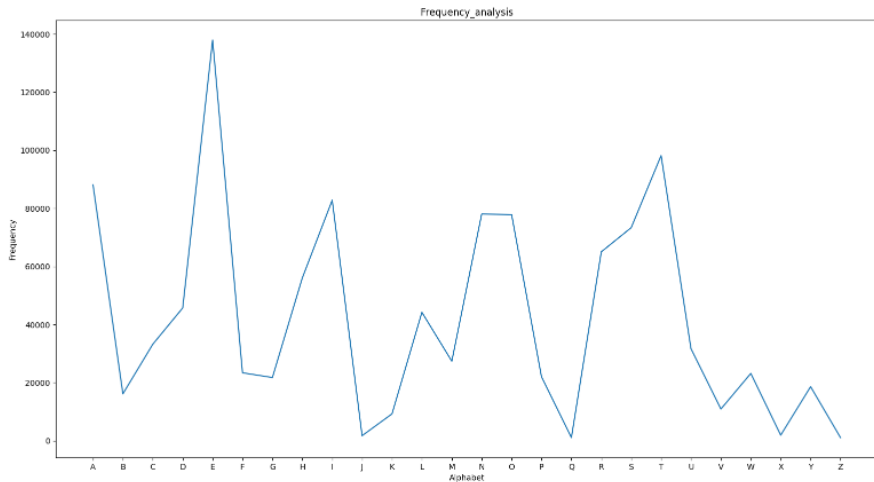
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Standard English Text Letter Statistics



Word Level Statistics

The most frequent English word is **the** ($\approx 6.421\%$). Attackers use word maps alongside single-letter distributions to quickly isolate key values.

Word	Frequency	Word	Frequency	Word	Frequency
the	6.421%	a	2.092%	i	0.945%
of	4.028%	in	1.778%	it	0.930%
and	3.150%	that	1.244%	for	0.770%
to	2.367%	is	1.034%	as	0.764%

Example Ciphertext

ZYPNM BEYQN DPCUM DSUMK XPBWU
SZKYS ZBTTK GCZQM CBGBI SYCGU
SZKYS ZDQGG SRUCG MDSNY SUSGZ
SBWQX ISYUQ YCQRM DCYXN QYMCU
LDBMY PMBCG MSZYS NMXQM Q

Ciphertext Letter Distribution Counters:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	8	9	6	1	0	8	0	2	0	4	1	13	5	0	4	9	2	15	2	9	0	2	4	12	8

Vigenère Cipher

- ▶ This is a **Polyalphabetic** substitution cipher
- ▶ The **keyword** determines the shift amount: $a \rightarrow 0$, $b \rightarrow 1$, ..., $z \rightarrow 25$.
- ▶ The keyword is **repeated** until it matches the plaintext length.
- ▶ Considered unbreakable in the 19th century; still used up through WWI

Example

Plaintext	s	e	c	r	e	t	m	e	s	s	a	g	i	n	g
Keyword	c	o	d	e	c	o	d	e	c	o	d	e	c	o	d
Shift	2	14	3	4	2	14	3	4	2	14	3	4	2	14	3
Ciphertext	U	S	F	V	G	H	P	I	U	G	D	K	K	B	J

Unlike Caesar's cipher, the shift changes periodically.

An Overview of Breaking Vigenère (Kasiski's Method)

Step 1: Isolate Key Length

- ▶ Look for **repeated sequences** within the ciphertext.
- ▶ The numerical distance between identical patterns will be a multiple of the keyword length ℓ .

Step 2: Isolate Individual Key Letters

Once the length ℓ is found, separate the ciphertext into ℓ isolated slices or buckets. Solve for each individual slice one by one using a standard frequency attack.

Kasiski's Method: Example

KZRWV GOZTW JQPIL WNSGM WHPAU XHYVS VBKVN WQVLL EMZMV VOECV BJCLQ
WGMQJ KMGAW JQMLI GXWZX AJWQV OLPLQ WGVHE WLXYP UWUIB VWZAI LQHTM
KLNAM LHUED IALPE STMGJ TQXBB VPLMK AUKIC MLAQH

- ▶ Assume the guessed key length is x .
- ▶ Letters in positions $0, x, 2x, \dots$ are encrypted using the same shift.
- ▶ Likewise, positions $1, x+1, 2x+1, \dots$ share another shift, and so on.
- ▶ Hence, each column forms a Caesar cipher with an individual shift (or key).

Len: 4
KZRW
VGJZ
TWJQ
PILW
NSGM
:

Len: 5
KZRWV
GOZTW
JQPIL
WNSGM
WHPAU
:

Kasiski's Method: Example

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Len:

4

KZRW

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TWJQ

PILW

NSGM

:

Len:

5

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JQPIL

WNSGM

WHPAU

:

Kasiski's Method: Estimating the Key Length

The **Index of Coincidence (IoC)** measures how likely two randomly selected letters from a text are the same. If the ciphertext contains N letters and the letter i appears n_i times, then

$$\mathcal{I} = \frac{26 \sum_{i=0}^{25} n_i(n_i - 1)}{N(N - 1)}.$$

Typical values are

Random text : $\mathcal{I} \approx 1.0$

English text : $\mathcal{I} \approx 1.7$

Kasiski's Method: Estimating the Key Length

Step 1: Find the Key Length

1. Guess a key length ℓ .
2. Split the ciphertext into ℓ slices.
3. Compute the IoC of each slice.
4. Average the IoCs of all slices.

The correct key length is the smallest ℓ whose average IoC is close to that of English.

C R Y P T O G R A P H Y
 k_1 k_2 k_3 k_4 k_1 k_2 k_3 k_4 k_1 k_2 k_3 k_4

Slice 1: C T A

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Slice 3: Y G H

Slice 4: P R Y

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k_1	k_2	k_3	k_4	k_1	k_2	k_3	k_4	k_1	k_2	k_3	k_4

Slice 1: C T A

Slice 2: R O P

Slice 3: Y G H

Slice 4: P R Y

Kasiski's Method: Finding the Key

Once the key length ℓ is known, each ciphertext slice is encrypted using a **single shift**. We recover the shift of each slice independently.

- ▶ Guess a shift $k \in \{0, \dots, 25\}$ and decrypt the slice.
- ▶ Let $\gamma_k = (\gamma_0, \dots, \gamma_{25})$ be the frequency vector of the decrypted slice.
- ▶ Let $\delta = (\delta_0, \dots, \delta_{25})$ be the frequency vector of English text.
- ▶ Measure the similarity between γ and δ using the **cosine similarity**

$$\cos(\theta) = \frac{\gamma \cdot \delta}{\sqrt{\gamma \cdot \gamma} \sqrt{\delta \cdot \delta}}, \quad \gamma \cdot \delta = \sum_{i=0}^{25} \gamma_i \delta_i.$$

- ▶ Repeat for all 26 possible shifts. The shift with the **largest cosine similarity** is chosen as the key letter for that slice.



One-Time Pad

definition

A **One-Time Pad** is a Vigenère cipher implementation where the key length is same as the plaintext length.

Question

Why is Kasiski's attack useless against a One-Time Pad?

One-Time Pad

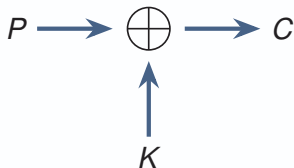
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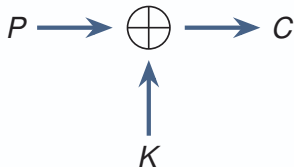
One-Time Pad Over Binary Alphabets



Example

Plaintext:	c (01100011)	y (01111001)	b (01100010)	e (01100101)	r (01110010)
Key String:	10101100	01100011	11010110	00111001	10110101
Cipher:	11001111	01011010	10110100	01011100	11000111

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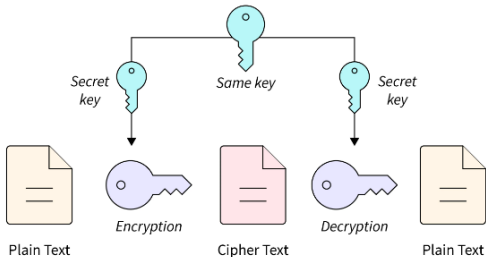
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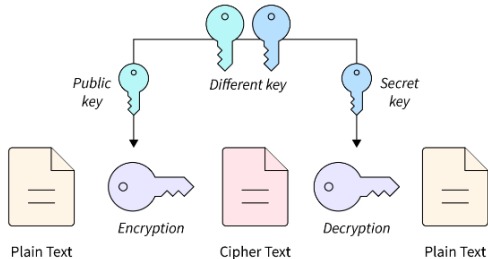
Modern Cryptography

Symmetric and Asymmetric Key Cryptography

Symmetric Encryption



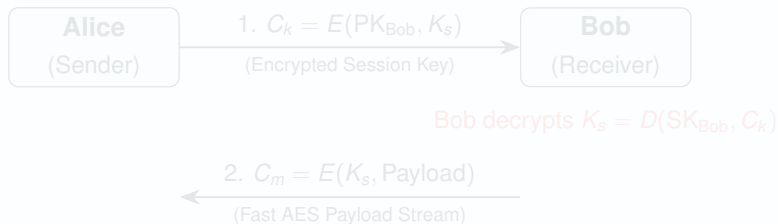
Asymmetric Encryption



Modern Crypto Infrastructure

The Core Principle

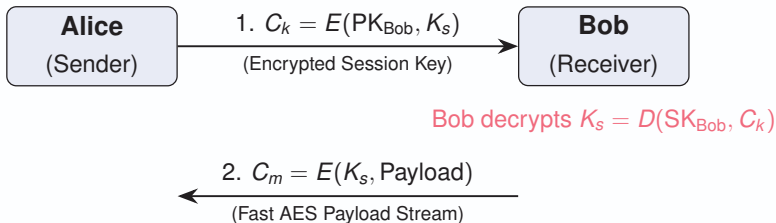
Symmetric-key cryptography is 'fast', but need a pre-shared secret key. Use **Asymmetric** cryptography to exchange a short-lived secret key, then use **Symmetric Encryption** to encrypt the actual payload data.



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Real-World Usage

- ▶ **HTTPS / TLS Handshake:** The foundation of secure web browsing.
- ▶ **SSH (Secure Shell):** Uses RSA/ECDSA to negotiate AES keys for terminal access.
- ▶ **PGP / GPG Email Security:** Encrypts large email attachments with symmetric keys while wrapping the key asymmetrically.

Key Takeaways

- ▶ Asymmetric crypto provides **key establishment**
- ▶ Symmetric crypto provides **efficient data transfer**

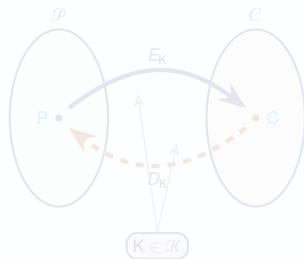
Symmetric Key Cryptosystem

Formal Definition: Symmetric Key Cryptosystem (Cipher)

A symmetric key cryptosystem is five-tuple:

$$(\mathcal{P}, \mathcal{C}, \mathcal{K}, \mathcal{E}, \mathcal{D})$$

- ▶ $\mathcal{P}$ is a finite set of possible **plaintexts**.
- ▶ $\mathcal{C}$ is a finite set of possible **ciphertexts**.
- ▶ $\mathcal{K}$ is a finite set of possible **keys**.
- ▶ $\mathcal{E}$ and $\mathcal{D}$ are set of deterministic algorithms



For each $K \in \mathcal{K}$, $\exists$ an **encryption** rule $E_K \in \mathcal{E}$ and a **decryption** rule $D_K \in \mathcal{D}$:

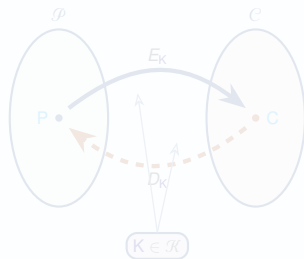
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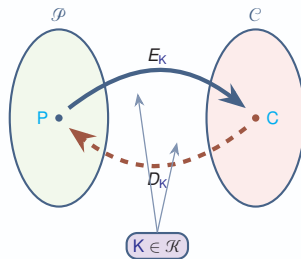
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Both algorithm sets $\mathcal{E}$ and $\mathcal{D}$ are publicly known

- ▶ Assume the attacker knows every structural feature and limit of the cipher
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Cipher Design Criteria

1. Must be **efficient** to encrypt and decrypt whenever the correct key is provided
2. Must be computationally **infeasible** to:
 - ▷ Decrypt data without knowing the key
 - ▷ Extract the key
 - ▷ Extrapolate any statistical information on the plaintext

One-Way Functions

These functions are public algorithms that are computationally **efficient** to compute forward, but exceptionally **difficult** to invert (without the key).

Cipher Design Criteria

1. Must be **efficient** to encrypt and decrypt whenever the correct key is provided
2. Must be computationally **infeasible** to:
 - ▷ Decrypt data without knowing the key
 - ▷ Extract the key
 - ▷ Extrapolate any statistical information on the plaintext

One-Way Functions

These functions are public algorithms that are computationally **efficient** to compute forward, but exceptionally **difficult** to invert (without the key).



Thank You for your
attention!

Any questions?